

Microeconomics

Lecture 7: Returns to Scale, Isoquants and MRTS

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A.Y. 2025–2026

6 March 2026

What we know so far

- Technology: $Q = f(L, K)$
- Marginal products: MPL , MPK
- Short run: diminishing marginal returns (with one fixed input)

Next step

What happens when **both** inputs increase? And how do we represent technology in the (L, K) plane using **isoquants**?

Learning objectives

After this lecture you will be able to:

1. Define **returns to scale**
2. Link returns to scale to economies/diseconomies of scale
3. Define **isoquants** and describe their properties
4. Compute and interpret the **MRTS**
5. Solve exercises on isoquants, MRTS and returns to scale

Focus of the lecture

Understanding **scale** and **substitutability**: how worthwhile it is to grow, and how easily you can substitute capital for labor.

Returns to scale: definition

Take $t > 1$ and compare:

$$f(tL, tK) \quad \text{with} \quad tf(L, K)$$

- ▶ **Increasing returns to scale:** $f(tL, tK) > tf(L, K)$
- ▶ **Constant returns to scale:** $f(tL, tK) = tf(L, K)$
- ▶ **Decreasing returns to scale:** $f(tL, tK) < tf(L, K)$

Intuition: what are we comparing?

If I double all inputs:

$$(L, K) \rightarrow (2L, 2K)$$

by how much does output increase?

- If output **more than doubles** $\Rightarrow$ increasing returns
- If output **exactly doubles** $\Rightarrow$ constant returns
- If output **less than doubles** $\Rightarrow$ decreasing returns

Caution

This is different from **diminishing marginal returns**: there *only one input* increased, here *all of them* do.

Comparison: marginal returns and returns to scale

Concept	Question
Marginal product	If I increase only L by 1, how much does Q grow?
Returns to scale	If I increase all inputs proportionally, how much does Q grow?

Message

These are two distinct concepts:

- you can have diminishing MPL
- and at the same time constant or increasing returns to scale

Cobb–Douglas and returns to scale

Given:

$$Q = AL^{\alpha}K^{\beta}$$

we can compute:

$$f(tL, tK) = A(tL)^{\alpha}(tK)^{\beta} = t^{\alpha+\beta}AL^{\alpha}K^{\beta}$$

which simplifies to:

$$f(tL, tK) = t^{\alpha+\beta}f(L, K)$$

So we simply need to look at the values of α and β :

- $\alpha + \beta > 1 \Rightarrow$ **increasing** RTS
- $\alpha + \beta = 1 \Rightarrow$ **constant** RTS
- $\alpha + \beta < 1 \Rightarrow$ **decreasing** RTS

Exercise

Classify the returns to scale of:

1. $Q = L^{0.7} K^{0.4}$

2. $Q = L^{0.5} K^{0.5}$

3. $Q = L^{0.3} K^{0.4}$

Solution

Sum of exponents:

1. $0.7 + 0.4 = 1.1 > 1 \Rightarrow$ increasing RTS
2. $0.5 + 0.5 = 1 \Rightarrow$ constant RTS
3. $0.3 + 0.4 = 0.7 < 1 \Rightarrow$ decreasing RTS

Why do economies of scale exist?

- **Specialization**: tasks more divided and more efficient
- **Indivisibilities**: large plants used better
- **Fixed costs**: spread over more output
- **Logistics / purchasing**: greater bargaining power

Example

A small bakery and a large industrial plant do not have the same efficiency if both double in size.

Intuition: increasing economies of scale

Case: e-commerce platform

An e-commerce firm that doubles its warehouses, software and staff does not always merely double its deliveries: it can achieve **more than twice as many** thanks to:

- standardized processes,
- software already developed,
- fixed costs spread over many orders,
- cheaper logistics purchasing.

Message

Economies of scale arise when growing in size makes the organization **more efficient**, not just bigger.

Why do diseconomies of scale emerge?

When the firm grows too large:

- more hierarchical layers
- more bureaucracy
- higher coordination costs
- more congestion of shared resources

Intuition

Bigger does not always mean more efficient.

Intuition: diseconomies of scale

Case: very large, multi-layered firm

When the firm grows too large, the following may increase:

- decision-making times,
- coordination costs,
- communication errors,
- conflicts between departments,
- bureaucratic rigidity.

Message

Beyond a certain size, organizational complexity can raise costs and reduce efficiency.

Exercise: returns to scale

A catering firm doubles:

$$L \rightarrow 2L, \quad K \rightarrow 2K$$

and observes that production rises from 1,000 meals to 2,400 meals.

1. What type of returns to scale do you observe?
2. Give a possible business explanation.

Solution

If output rises from 1,000 to 2,400, it more than doubles:

$$2,400 > 2 \cdot 1,000$$

So there are **increasing returns to scale**.

Possible explanation

The central kitchen, logistics and organization are used more effectively at a larger scale.

From the output level to the input map

So far we have written:

$$Q = f(L, K)$$

But we can also fix an output level $\bar{Q}$ and ask:

Question

Which combinations of labor and capital produce exactly the same output?

Isoquants: definition

An **isoquant** is the set of combinations (L, K) that produce the same output $\bar{Q}$:

$$\{(L, K) : f(L, K) = \bar{Q}\}$$

Analogy

Isoquants are the analogue of indifference curves, but here they represent the **same output**, not the same utility.

What does an isoquant tell us?

Managerial reading

An isoquant shows all the combinations of labor and capital that yield the **same level of output**.

Case: warehouse

To handle 10,000 orders/month you can:

- use more workers and less automation,
- or fewer workers and more technology.

Both combinations can lie on the same isoquant.

Why are isoquants useful?

They allow us to visualize:

- how much you can substitute K for L
- whether the technology is rigid or flexible
- how the optimal use of inputs changes when prices change

Typical properties of isoquants

With a standard technology:

- negative slope
- they do not intersect
- convex to the origin

Intuition

If you have more labor, you can generally use less capital to obtain the same output.

Why is the slope negative?

If you want to keep Q constant:

- if you increase L , you can reduce K
- if you reduce L , you must increase K

Example

In a warehouse, more automation can reduce labor requirements; more workers can make up for less automation.

Why do isoquants not intersect?

If two isoquants intersected at the same point (L, K) , that same combination would produce two different output levels.

$$f(L, K) = Q_1 \quad \text{and} \quad f(L, K) = Q_2, \quad Q_1 \neq Q_2$$

Contradiction

The production function assigns a single output to each input combination.

Why are isoquants convex?

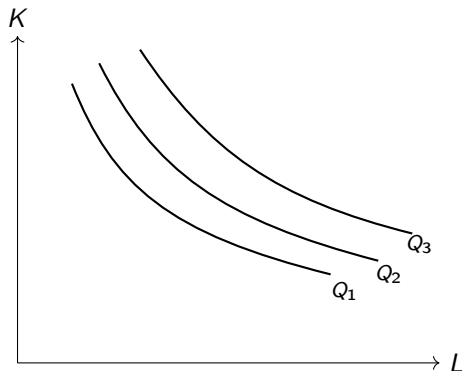
Convexity reflects a standard idea:

- when you already use a lot of labor and little capital,
- substituting further labor for capital becomes increasingly difficult.

Message

Technical substitution is possible, but neither unlimited nor at a constant rate.

Graph: isoquants for increasing output



Reading

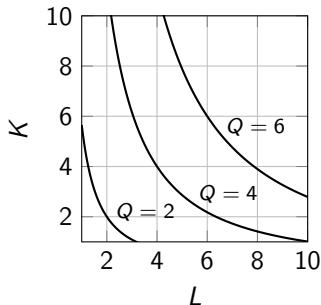
Isoquants farther from the origin correspond to higher output.

Graph: Cobb–Douglas isoquants

Consider:

$$Q = L^{0.6} K^{0.4}$$

Isoquants for three output levels.



What we see

Isoquants are downward-sloping and convex: the trade-off between L and K is not constant.

How to read the graph of Cobb–Douglas isoquants

- Isoquants farther from the origin correspond to higher output
- If you increase L , you can reduce K while keeping the same output
- But the possible reduction in K becomes progressively smaller

Intuition

If you already have a lot of labor and little capital, one more worker cannot replace much capital: the MRTS diminishes.

MRTS: definition

Along an isoquant, $dQ = 0$:

$$dQ = MPL dL + MPK dK = 0$$

Hence:

$$-\frac{dK}{dL} \Big|_{Q=\bar{Q}} = \frac{MPL}{MPK} \equiv MRTS_{LK}$$

Interpretation

How much capital you can give up if you increase labor by 1 unit while keeping output constant.

Economic interpretation of the MRTS

If $MRTS = 2$, this means that:

- by increasing L by 1,
- you can reduce K by about 2
- without changing Q

Note

The MRTS depends on the point where you are on the isoquant.

Why does the MRTS usually diminish?

If you already have a lot of labor and little capital:

- one more worker is of relatively little use
- giving up more capital becomes costly in terms of output

Message

The amount of capital you can “replace” with labor diminishes as you use more labor.

MRTS in practice

Case: supermarket

To run the checkouts you can combine:

- more cashiers,
- more self-checkout machines.

Reading the MRTS

The MRTS tells you how many self-checkout machines you can “save” if you add one cashier, keeping the service level constant.

Message

The more labor-intensive you already are, the less one more cashier can replace technology.

Exercise

Given:

$$Q = L^{0.6} K^{0.4}$$

1. Compute MPL and MPK
2. Compute $MRTS = \frac{MPL}{MPK}$
3. Evaluate the MRTS at $(L, K) = (16, 9)$

Solution

$$MPL = 0.6L^{-0.4}K^{0.4} \quad MPK = 0.4L^{0.6}K^{-0.6}$$

$$MRTS = \frac{MPL}{MPK} = \frac{0.6}{0.4} \cdot \frac{K}{L} = 1.5 \cdot \frac{K}{L}$$

At (16, 9):

$$MRTS = 1.5 \cdot \frac{9}{16} = 0.84375$$

Interpretation

If I increase L by 1, I can reduce K by about 0.84 while keeping Q constant.

Exercise

At a certain point on the isoquant, the MRTS is:

$$MRTS = 3$$

1. How do you interpret this number?
2. If the MRTS falls from 3 to 1.5 along the same isoquant, what does this mean economically?

Solution

- 1) If $MRTS = 3$, by increasing L by 1 unit you can reduce K by about 3 units while keeping Q constant.
- 2) If the $MRTS$ falls from 3 to 1.5, labor becomes less able to replace capital: you are already using a lot of labor relative to capital.

Three cases of substitutability

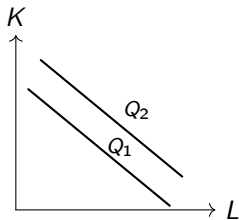
- **Perfect substitutes:** linear isoquants
- **Perfect complements:** L-shaped isoquants
- **Cobb–Douglas / standard case:** convex isoquants

Why does it matter?

Because the shape of the isoquants tells us how easy it is to reorganize production.

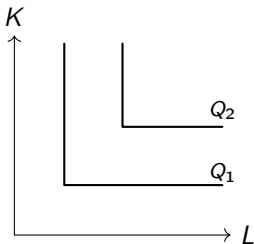
Three shapes of isoquants

Perfect substitutes



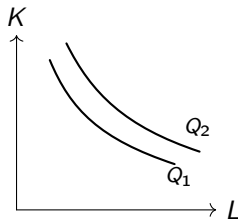
Constant trade-off between L and K

Perfect complements



Fixed proportions required

Standard case



Partial substitutability, diminishing
MRTS

Perfect substitutes

Intuitive example

Two identical workers performing exactly the same standardized task.

If 1 unit of K can always be replaced by 1 unit of L at a constant rate:

- linear isoquants
- constant MRTS

Perfect complements

Intuitive example

1 driver + 1 truck.

If fixed proportions are required:

- adding only one of the two inputs is useless
- right-angled isoquants

Message

Inputs do not always substitute for each other: sometimes they **combine**.

Standard case

Most real-world technologies lie between the two extremes:

- substitutability exists
- but it is not perfect
- and the MRTS diminishes

Example

More automation can reduce labor, but not eliminate it entirely.

Exercise

- Explain in 4 lines why two isoquants cannot intersect.

If two isoquants intersected at the same point (L, K) , the same input combination would produce two different output levels.

This contradicts the definition of a production function, which assigns a single output to each input combination.

Exercise

Given:

$$Q = L^{0.6}K^{0.4}$$

compute the MRTS at:

$$(L, K) = (20, 10)$$

Solution

$$MRTS = \frac{0.6}{0.4} \cdot \frac{K}{L} = 1.5 \cdot \frac{K}{L}$$

$$MRTS = 1.5 \cdot \frac{10}{20} = 0.75$$

Interpretation

+1 unit of labor allows a reduction of about 0.75 units of capital, keeping output constant.

Exercise

For each case, state whether the inputs are closer to:

- perfect substitutes
- perfect complements
- an intermediate case

1. driver and truck
2. human cashiers and self-checkout machines
3. two identical clerks entering standardized data

Solution

1. **Driver and truck:** perfect complements.
2. **Cashiers and self-checkout machines:** intermediate case, with some substitutability but not perfect.
3. **Two identical clerks:** closer to perfect substitutes.

Group activity

Work in groups.

Choose a real firm and discuss:

1. where might there be economies of scale?
2. where might diseconomies emerge?
3. are the inputs closer to substitutes or to complements?

Why do we need all this?

Minimizing costs

The firm does not just want to produce: it wants to produce **at minimum cost**.

To do so it will have to combine:

- **isoquants**: technology
- **isocosts**: input prices

Idea

The optimal choice of inputs will arise from the meeting of: the *technical trade-off* and the *cost trade-off*

Overall summary

- Returns to scale: comparison between $f(tL, tK)$ and $tf(L, K)$
- Isoquants: input combinations that produce the same output
- MRTS: technical trade-off between labor and capital
- Three cases: substitutes, complements, intermediate

Skills acquired

- ✓ Classify returns to scale
- ✓ Read and draw isoquants
- ✓ Compute and interpret the MRTS
- ✓ Apply the concepts to business examples

Optional homework exercises

1. For $Q = L^{0.4}K^{0.6}$ compute the MRTS and comment on how it varies as K/L changes.
2. Explain why isoquants are typically convex.
3. List two economies and two diseconomies of scale for a firm you know.